DATA SCIENCE & MACHINE LEARNING

MODULE 1

DATA SCIENCE

Data Science is an interdisciplinary field that deals with the collection, processing, analysis, and interpretation of large volumes of data to extract useful insights and support decision-making.

- It integrates concepts from statistics, computer science, mathematics and domain knowledge.

In simple terms, Data Science answers questions such as:

- What happened? (Descriptive Analysis)
- Why did it happen? (Diagnostic Analysis)
- What will happen next? (Predictive Analysis)
- What should be done? (Prescriptive Analysis)

Evolution of Data Science

Data Science has evolved from multiple fields:

- **Statistics** → For data analysis and inference
- **Database Systems** → For storing and managing data
- **Artificial Intelligence (AI)** → For intelligent decision-making
- **Big Data Technologies** → For handling large-scale data

With the growth of the internet, IoT, and digital systems, the volume of data has increased exponentially, making Data Science essential.

Tools and Technologies

- **Programming Languages**
  - Python (most popular)
  - R
- **Libraries and Frameworks**
  - NumPy, Pandas (data handling)
  - Scikit-learn (machine learning)
  - TensorFlow, Keras (deep learning)
- **Data Visualization Tools**
  - Matplotlib
  - Seaborn
  - Tableau
  - Power BI
- **Databases**
  - SQL (MySQL, PostgreSQL)
  - NoSQL (MongoDB)

Real-Life Example of Data Science: Supermarket

- Imagine a large supermarket

Problem

The supermarket wants to:

- Know which products customers buy the most.
- Predict future sales.
- Reduce food waste.
- Offer personalized discounts.

How Data Science Helps

Step	Real-Life Task	What the Programmer Does	Python Library Used
1. Collect Data	Customers buy products.	Read sales data from a CSV file or database.	Pandas
2.Clean Data	Some records have missing or duplicate values.	Remove duplicates, fill missing values, fix incorrect data.	Pandas
3. Analyze Data	Find the best-selling products.	Count how many times each product was sold.	Pandas, NumPy
4.Visualize Data	Show sales in graphs.	Create bar charts and pie charts.	Matplotlib, Seaborn
5.Train Model	Predict next month's sales.	Split data into training and testing sets, then train a model.	Scikit-learn
6.Test Model	Check prediction accuracy.	Compare predicted values with actual values.	Scikit-learn
7.Predict	Estimate future sales.	Input new data and get predicted sales.	Scikit-learn
8.Business Decision	Order more products.	Use predictions to make inventory decisions.	AI/Business Logic

Simple Scenario

Suppose the supermarket notices:

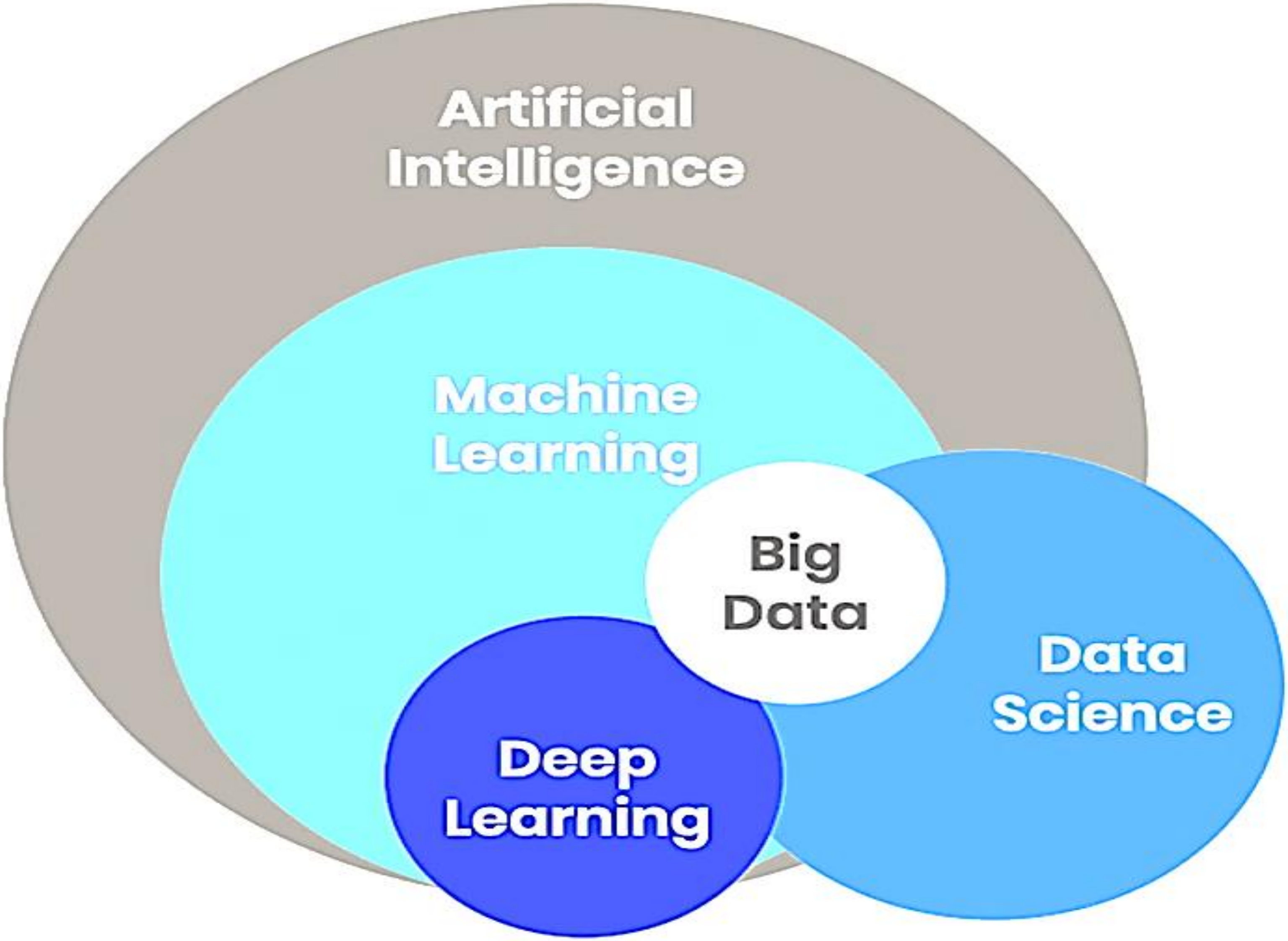
Customer	Items Bought
A	*Bread, Milk*
B	*Bread, Eggs*
C	*Bread, Milk*
D	*Bread, Milk, Butter*

A **data scientist** analyzes this data and discovers:

- Bread is the **best-selling product**.
- Customers who buy **bread usually buy milk**.
- The supermarket places bread and milk closer together to increase sales.

Result

✔Higher sales
✔Better customer satisfaction
✔Less food waste
✔Smarter business decisions


Artificial Intelligence
Machine Learning
Big Data
Data Science
Deep Learning

Artificial Intelligence (AI) – Parent field

- This is the **largest circle**, meaning it is the **broadest field**.
- AI refers to machines that can simulate human intelligence (thinking, reasoning, decision-making).
- Examples: chatbots, self-driving cars, virtual assistants.
- So, **everything inside comes under AI**.

Machine Learning (ML)

- ML is a **subset of AI**.
- It allows machines to **learn from data automatically** without explicit programming.
- Instead of coding rules, we train models using data.
- Example: Email spam detection, recommendation systems.

Deep Learning (DL)

- DL is a **subset of Machine Learning**.
- It uses **neural networks** (similar to human brain structure).
- Works well for complex tasks.
- Example: Image recognition, speech recognition.

- **Big Data**
  - Big Data refers to **huge volumes of data** (structured + unstructured).
  - It overlaps with ML and Data Science because:
    - ML needs large data to learn
    - Data Science uses big data for analysis
  - Example: Social media data, sensor data, online transactions.
- **Data Science**
  - Data Science overlaps with Big Data and partially with ML.
  - It includes:
    - Data collection
    - Data analysis
    - Visualization
    - Some ML techniques
  - *Data Science is not fully inside AI*
    - Because it also involves statistics, analysis, and decision-making (not only AI)

- **Big Data** = Lots of data
- **Data Science** = Understands the data
- **Machine Learning** = Learns from the data
- **Deep Learning** = Learns complex patterns (like images and speech)
- **Artificial Intelligence** = Makes smart decisions

Think of **a self-driving car**:

- **AI** = The whole self-driving car.
- **Machine Learning** = Learns how to drive from past driving data.
- **Deep Learning** = Recognizes traffic signs, pedestrians, and other vehicles using cameras.
- **Data Science** = Analyzes driving data to improve the car's performance.
- **Big Data** = Millions of miles of driving data collected from many cars.

DATA SCIENCE CLASSIFICATION

Data Science can be classified based on

- **Type of Analytics**
- **Type of data**
- **Techniques used**
- **Application areas**.

Understanding this classification helps in choosing the right method for solving real-world problems.

<u>Type of Analytics</u>

Descriptive Analytics

- **Question Answered**: *What happened?*
- **Purpose**: To summarize historical data
- **Techniques Used**:
  - Aggregation (sum, average)
  - Data visualization (charts, graphs)

Examples:

- Monthly sales report
- Student performance summary

Tools:

- Excel, Tableau, Power BI
- It gives a **clear picture of past events** but does not explain causes.

Diagnostic Analytics

- **Question Answered**: *Why did it happen?*
- **Purpose**: To identify root causes of a problem
- **Techniques Used**:
  - Drill-down analysis
  - Correlation analysis
  - Data Mining
- **Examples:**
  - Why did crop production decrease?
  - Why did a company face loss?
- It helps in **understanding reasons behind outcomes**.

Predictive Analytics

- **Question Answered**: *What will happen?*
- **Purpose**: To forecast future outcomes
- **Techniques Used**:
  - Regression
  - Classification
  - Time series analysis
  - Machine Learning algorithms
- **Examples:**
  - Weather forecasting
  - Stock price prediction
  - Farmer risk prediction

  It uses past data to **predict future trends**.

Prescriptive Analytics

- **Question Answered**: *What should be done?*
- **Purpose**: To recommend actions
- **Techniques Used**:
  - Optimization algorithms
  - Simulation models
  - Artificial Intelligence
- **Examples:**
  - Suggesting best crop for farmers
  - Route optimization in delivery systems

  It provides **decision-making solutions**.

Type of Analytics	Question Answered	Example
Descriptive → Past	What happened?	Sales report
Diagnostic → Reason	Why happened?	Reason for loss
Predictive → Future	What will happen?	Weather prediction
Prescriptive → Action	What to do?	Recommendation system

Type of Data

- Structured Data
- Unstructured Data
- Semi-Structured Data

Structured data

- Data that is organized in a predefined format, making it easy to store, manage, search, and analyze.
- It follows a fixed structure (schema) where every piece of information has a specific place.
- Structured data is data that is arranged in **rows and columns**, similar to a table.
- Each row represents a single record, while each column represents a specific attribute or field

Student ID	Name	Marks
101	Rakhil	85
102	Priya	90
103	Aman	78

Here:

Each row represents one student, Each column stores a specific type of information.

Unstructured Data

- Unstructured data is data that does not have a predefined format or fixed schema.
- Unlike structured data, it is not organized into rows and columns.
- It consists of information in various forms such as text, images, audio, and videos, making it more difficult to store, process, and analyze using traditional databases.

Semi-Structured Data

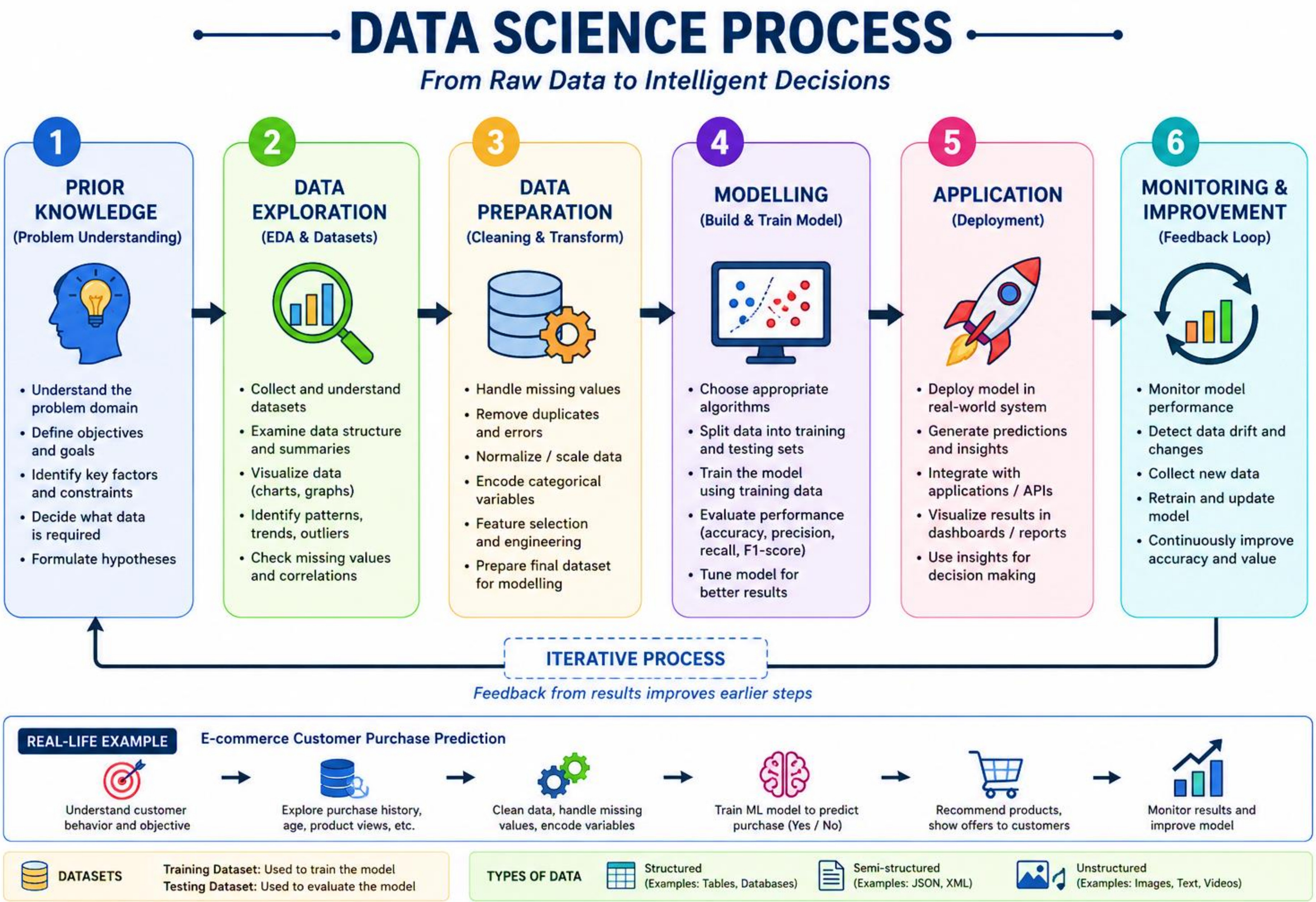
- Semi-structured data is data that is partially organized.
- It does not follow the strict row-and-column format of structured data but contains tags, labels, or a hierarchical structure that helps organize and identify the data.
- It lies between structured and unstructured data.

Application Areas

- **Business Analytics-**Used in marketing, sales, finance
- **Healthcare Analytics-**Disease prediction, patient monitoring
- **Agricultural Analytics-**Crop prediction, farmer risk analysis
- **Social Media Analytics-**Sentiment analysis, trend detection
- **Financial Analytics-**Fraud detection, risk management

Technique	Description	Key Concepts / Techniques	Applications / Uses
Statistical Analysis	Foundation of Data Science used to collect, analyze, interpret, and present data.	Mean, Median, Mode Variance & Standard Deviation Probability Distributions	Inference Hypothesis Testing
Machine Learning	Learns patterns from data to make predictions or decisions without explicit programming.	**Supervised Learning** – Learns using labeled data. **Unsupervised Learning** – Learns from unlabeled data by identifying hidden patterns or groups. **Reinforcement Learning** – Learns through trial and error using rewards and penalties.	Prediction Classification Recommendation Systems Fraud Detection
Deep Learning	Uses Artificial Neural Networks (ANNs) to solve complex problems.	Artificial Neural Networks (ANNs) Multiple hidden layers	Image Recognition Speech Processing
Data Mining	Extracts hidden patterns, relationships, and useful information from large datasets.	Clustering Association Rules Classification	Customer Segmentation Market Basket Analysis Fraud Detection
Big Data Analytics	Deals with storing and analyzing large-scale datasets that cannot be processed using traditional methods.	Hadoop Spark	Large-scale Data Processing Real-time Analytics, Business Intelligence

DATA SCIENCE PROCESS



1. Data Science Process Overview

- A structured approach to extract insights from data
- Converts **raw data → meaningful information → decision-making**
- Involves multiple steps: understanding, analyzing, modelling, and applying
- The process is **iterative** (can repeat for improvement)

2. Prior Knowledge (Problem Understanding)

- Clearly define the **problem statement**
- Identify **objectives and expected outcomes**
- Understand domain-specific factors (business, agriculture, healthcare, etc.)
- Decide:
  - What data is needed?
  - What type of analysis is required?
- Helps in selecting appropriate techniques and tools

3. Data Exploration (Exploratory Data Analysis – EDA)

- Initial investigation of data to understand:
  - Structure
  - Patterns
  - Relationships
- Works on **datasets** (collection of structured/unstructured data)

Activities:

- Summary statistics:
  - Mean, median, mode
  - Standard deviation
- Data visualization:
  - Bar charts, histograms, scatter plots
- Identifying:
  - Missing values
  - Outliers
  - Data distribution

Purpose:

- Detect errors and inconsistencies
- Understand important features
- Guide data preparation and modelling

4. Data Preparation (Preprocessing)

- Most **time-consuming and critical step**
- Converts raw data into clean and usable format

Key Steps:

- Data Cleaning:
  - Handle missing values (remove/replace)
  - Remove duplicates
- Data Transformation:
  - Normalization and scaling
  - Encoding categorical variables (label encoding, one-hot encoding)
- Feature Engineering:
  - Creating new features
  - Selecting important features
- Data Integration:
  - Combining data from multiple sources

Outcome:

- High-quality dataset ready for modelling

5. Modelling

- Building models using algorithms from Machine Learning

Types of Models:

- Supervised Learning
- Unsupervised Learning

Steps:

- Split data into training and testing sets
- Train the model using training data
- Test the model on unseen data

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1-score

Advanced Techniques:

- Deep Learning for complex tasks

6. Application (Deployment)

- Implementing the model in real-world systems
- Making predictions available to users

Examples:

- Web applications
- Mobile apps
- Decision support systems

Tasks:

- Integrating model with software
- Automating predictions
- Generating reports and dashboards

7. Monitoring and Maintenance

- Continuous evaluation of model performance
- Updating model with new data

Example

Prior Knowledge → Understand symptoms, medical history	Problem: Predict whether a customer will purchase a product Factors: age, browsing history, past purchases, product views
Data Exploration → Analyze patient records	Analyze which products are most viewed Identify customer buying patterns
Data Preparation → Clean missing health data	Convert "Yes/No" purchase into 1/0 Normalize customer income values
Modelling → Apply classification model	Apply classification model
Application → Predict disease (Yes/No)	E-commerce website recommends products Predicts and shows "Customers like you also bought…"
Monitoring → Improve model with new patient data	Improve model with new data

DATASETS

- A **dataset** is a collection of data organized in a structured, semi-structured, or unstructured format.
- It is used for data analysis, training machine learning models, and decision-making.
- A dataset serves as the input for any data science or machine learning task.

1. Structure of a Dataset

- Most datasets are represented in **tabular form**.

Feature 1	Feature 2	Feature 3	Target
Value	Value	Value	Output

Components of a Dataset

Rows → Records or observations
Columns → Features or variables
Target Variable → Output value (used only in supervised learning)

Example

Weather	Pest	Finance	Insurance	Risk
High	Low	Medium	Yes	No Risk
Low	High	Low	No	Risk

In this example:

Features: Weather, Pest, Finance, Insurance

Target Variable: Risk

2. Types of Datasets

a) Structured Data
b) Unstructured Data
c) Semi-Structured Data

3. Dataset Characteristics

a) Size

- **Small dataset:** Faster to process but may produce less accurate results.
- **Large dataset:** Improves learning but requires more computational resources.

b) Quality

- **Clean data:** Produces better model performance.
- **Noisy data:** Contains errors or inconsistencies, leading to poor performance.

c) Balance

- **Balanced dataset:** Contains approximately equal numbers of samples in each class.
- **Imbalanced dataset:** One class has significantly more samples than the others.

Example:
If most farmers are classified as **"No Risk"**, the model may become biased toward predicting **"No Risk"** for new cases.

4. Dataset Operations

a) Data Cleaning

Data cleaning improves data quality by:

- Removing missing values
- Correcting errors and inconsistencies
- Eliminating duplicate records

b) Data Transformation

- Data transformation converts data into a suitable format for machine learning.

Normalization

- Normalization is the process of scaling numerical values into a fixed range, usually **0 to 1**, so that all features contribute equally to the model.

Encoding

- Encoding converts categorical (text) data into numerical values because machine learning algorithms work only with numerical data.

c) Data Splitting

Data splitting is the process of dividing a dataset into different subsets for training, validation, and testing. It helps ensure that the model performs well on unseen data.

Why Do We Split Data?

- If a model is trained and tested using the same data, it may simply memorize the data instead of learning general patterns. This problem is called **overfitting**.

Data splitting helps to:

- – Measure the model's real performance
- – Detect overfitting
- – Tune model parameters effectively
- – Improve generalization to new data

Common Types of Data Splits

1. Training Set

- Used to train the model.
- Typically contains **60–80%** of the dataset.
- The model learns patterns from this data.

2. Validation Set

- Used to tune hyperparameters and select the best model.
- Typically contains **10–20%** of the dataset.
- Helps prevent overfitting.

3. Test Set

- Used only after training is complete.
- Typically contains **10–20%** of the dataset.
- Provides an unbiased evaluation of the final model.

Example of Data Splitting

- Suppose a dataset contains **1000 samples**.

Dataset	Number of Samples
Training Set	700
Validation Set	150
Test Set	150

This represents a **70% : 15% : 15%** data split, which is commonly used in machine learning.

DESCRIPTIVE STATISTICS

1. UNIVARIATE DATA ANALYSIS

a. MEASURES OF CENTRAL TENDENCY

Mean (Average)
- Sum of all values divided by count

$$\bar{x} = \frac{\sum x_i}{n}$$

Median (Middle value)
- Middle value when data is in order

2, 3, 4, 5, 6
4

Mode (Most frequent)
- Value that appears most often

2, 2, 3, 3, 3, 4
3

b. MEASURES OF DISPERSION

Range
- Max value − Min value

Max = 10, Min = 2
Range = 8

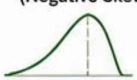
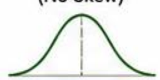
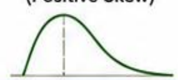
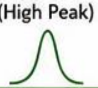
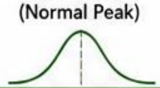
Variance
- Average of squared deviations from mean

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

Standard Deviation
- Square root of variance

$$s = \sqrt{s^2}$$

c. DATA DISTRIBUTION

Skewness	Left Skew (Negative Skew) 	Symmetrical (No Skew) 	Right Skew (Positive Skew) 
Kurtosis	Leptokurtic (High Peak) 	Mesokurtic (Normal Peak) 	Platykurtic (Flat Peak) 

d. VISUALIZATION

Histogram
(Continuous Data)

Bar Chart
(Categorical Data)

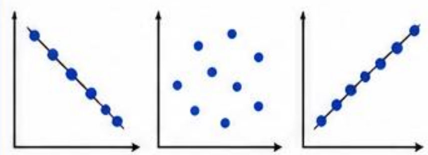
Box Plot
(Shows Distribution)

2. MULTIVARIATE DATA ANALYSIS

a. CORRELATION

- Measures strength and direction of linear relationship between two variables
- Correlation coefficient (r) ranges from −1 to +1

r = −1 Perfect Negative | r = 0 No Correlation | r = +1 Perfect Positive



b. COVARIANCE

- Measures the direction of the linear relationship between two variables

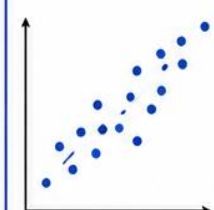
$$Cov(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

Covariance > 0
Variables move in the same direction

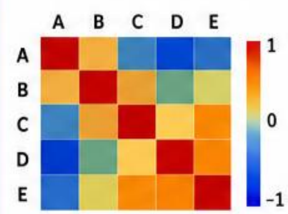
Covariance < 0
Variables move in opposite directions

c. MULTIVARIATE VISUALIZATION

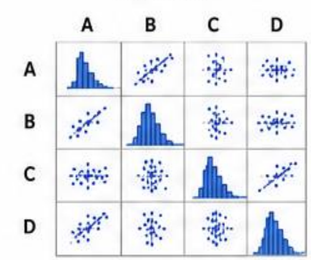
Scatter Plot
(Relationship between two variables)



Heatmap
(Correlation between many variables)



Pair Plot
(Pairwise relationships in dataset)



In Short: Descriptive Statistics helps summarize and describe the main features of data using numbers, tables and graphs.

Descriptive statistics are used to **summarize, organize, and describe the main features of data** in a simple and meaningful way.

1. Univariate Data Analysis

- **Univariate** means analyzing **only one variable at a time**.
- Understand the distribution and characteristics of a single feature.

Key Measures in Univariate Analysis

a) Measures of Central Tendency

These show the "center" of the data.

- **Mean (Average)**
- **Median (Middle value)**
- **Mode (Most frequent value)**

Example (Student Marks):
Marks = 40, 50, 60, 70, 80

- Mean = 60
- Median = 60
- Mode = None

b) Measures of Dispersion

These show how spread out the data is.

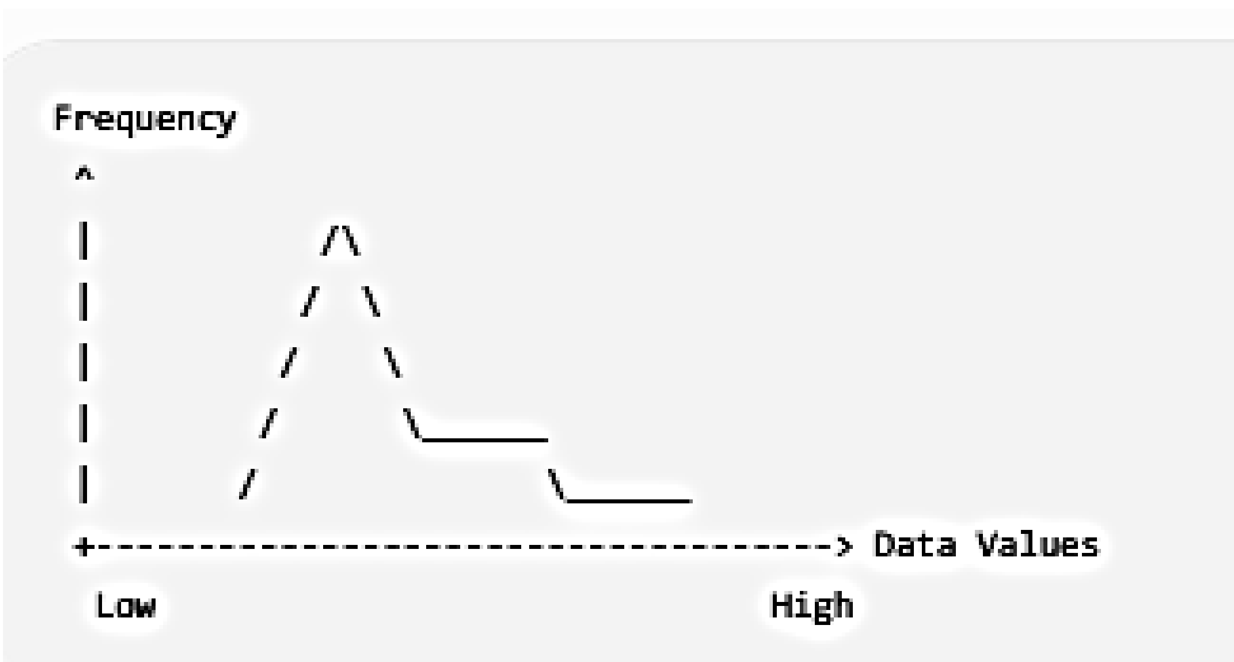
- **Range** = Max − Min
- **Variance** (It shows how far the data values are from the average.)
- **Standard Deviation** (**square root of variance**.)

c) Data Distribution

- 1. Positive Skew (Right Skew)
- 2. Negative Skew (Left Skew)
- 3. Symmetrical Distribution

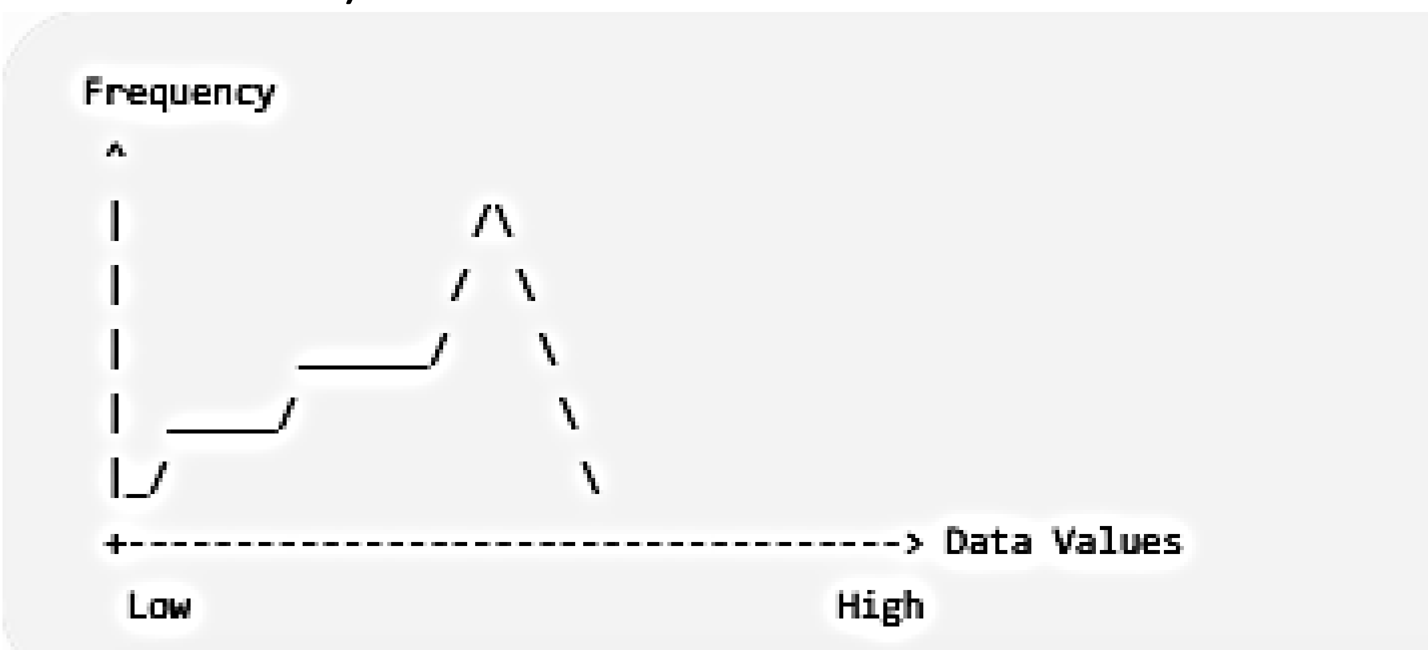
Positive Skew (Right Skew)

- **X-axis:** Data values (Low → High)
- **Y-axis:** Frequency
- **Tail:** Long tail on the **right**
- **Example:** Salaries (most people earn lower salaries, a few earn very high salaries)



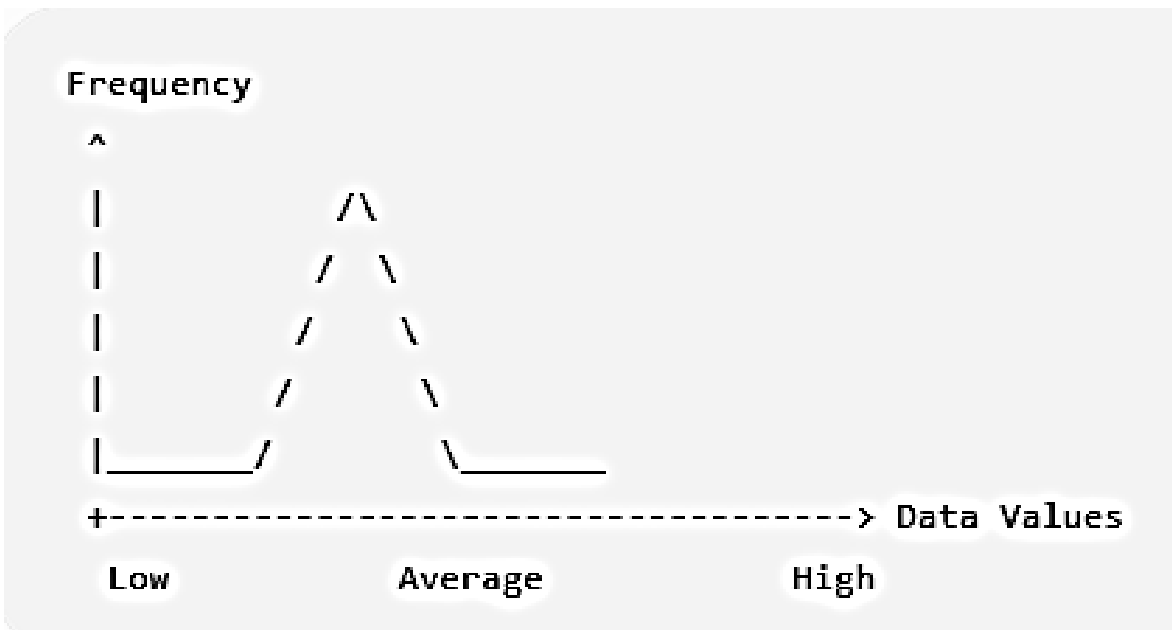
Negative Skew (Left Skew)

- **X-axis:** Data values (Low → High)
- **Y-axis:** Frequency
- **Tail:** Long tail on the **left**
- **Example:** Easy exam marks (most students score high, only a few score low)



Symmetrical Distribution

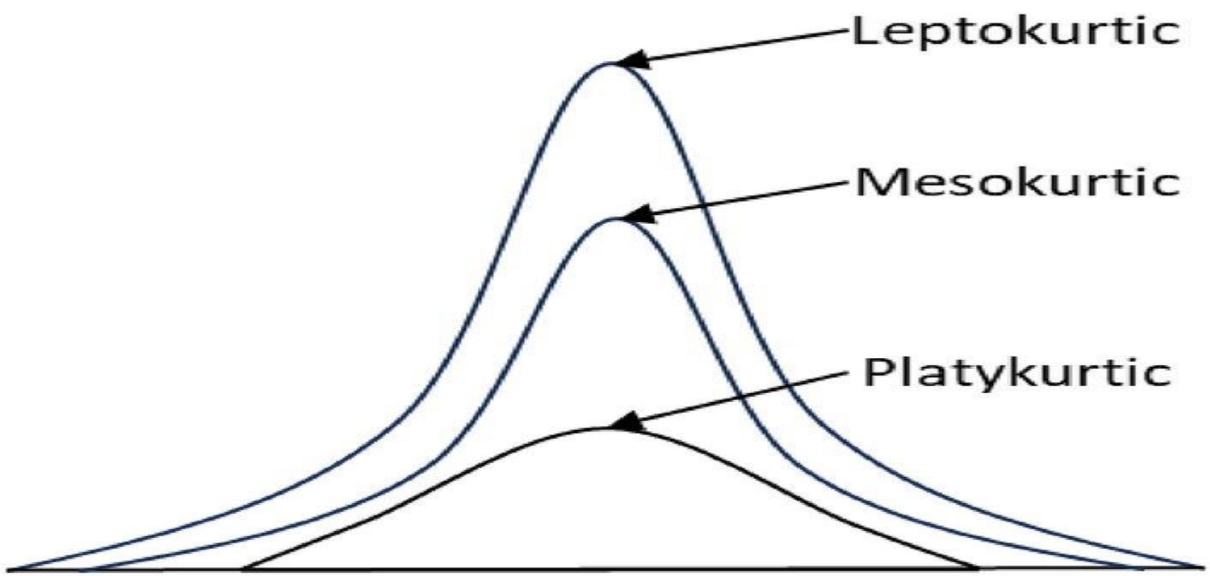
- **X-axis:** Data values
- **Y-axis:** Frequency
- Both sides are equal.
- **Example:** Heights of students.



Kurtosis

Kurtosis shows how high or flat the peak of the data is.

- **Leptokurtic:** High peak.
- **Mesokurtic:** Normal peak.
- **Platykurtic:** Flat peak.



d) Visualization

- Histogram , Bar chart , Box plot

2. Multivariate Data Analysis

- **Multivariate** means analysing **two or more variables together**.
- Understand **relationships and interactions** between variables.

Key Techniques in Multivariate Analysis

a) Correlation

  - Measures relationship between variables
  - Value ranges from **-1 to +1**
  - Example:
    - **Study Hours vs Marks**
    - **Attendance vs Marks**

b) Covariance- Shows direction of relationship

c) Multivariate Visualization

  - Scatter plots
  - Heatmaps
  - Pair plots

ASSIGNMENT

Question 1:

A dataset shows the monthly income (₹) of 8 individuals:
10,000, 12,000, 15,000, 18,000, 20,000, 22,000, 25,000, 30,000

- Calculate **Mean, Median, Mode**
- Find **Range, Variance, Standard Deviation**

Question 2:

The following data shows Experience (years) and Salary (₹):

Experience	Salary
1	15,000
3	25,000
5	40,000
7	55,000
9	70,000

- Find the **relationship** between variables
- Calculate **covariance**

Question 3:

The following data represents marks of 10 students:
12, 15, 18, 20, 22, 25, 28, 30, 32, 35

- Calculate **Mean**
- Find **Variance**
- Compute **Standard Deviation**

Question 4:

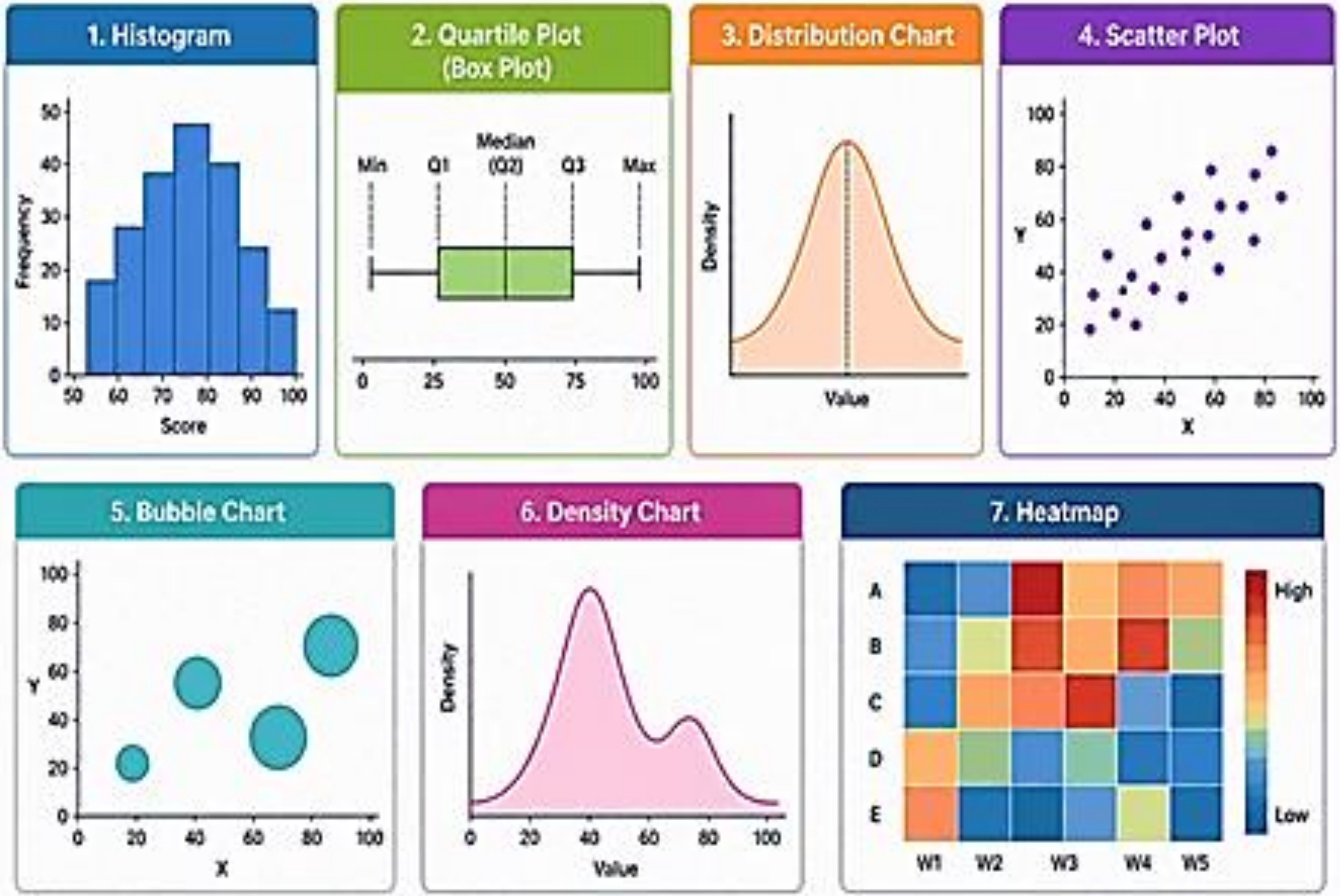
Mean = 20
Data: **10, 15, 25, x, 30**

- Find **x**

DATA VISUALIZATION

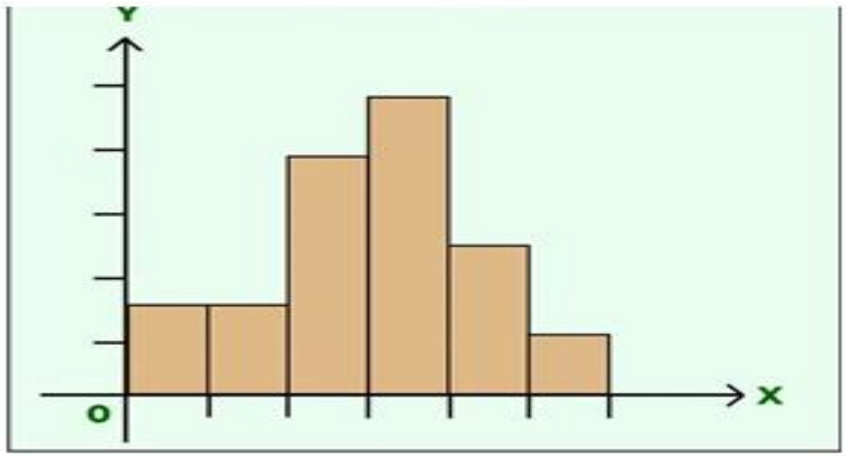
- **Data Visualization** is the graphical representation of data using charts, graphs, and plots.
- It converts raw numerical data into visual formats that are easy to understand.
- It helps users identify patterns, trends, relationships, and outliers quickly.
- It is widely used in business, research, education, healthcare, and data science.

TYPES OF DATA VISUALIZATION



Histogram

- A **Histogram** is a graphical representation of the **frequency distribution of continuous numerical data.** It shows how many data values fall within a particular range (called a **bin** or **class interval**).



Features

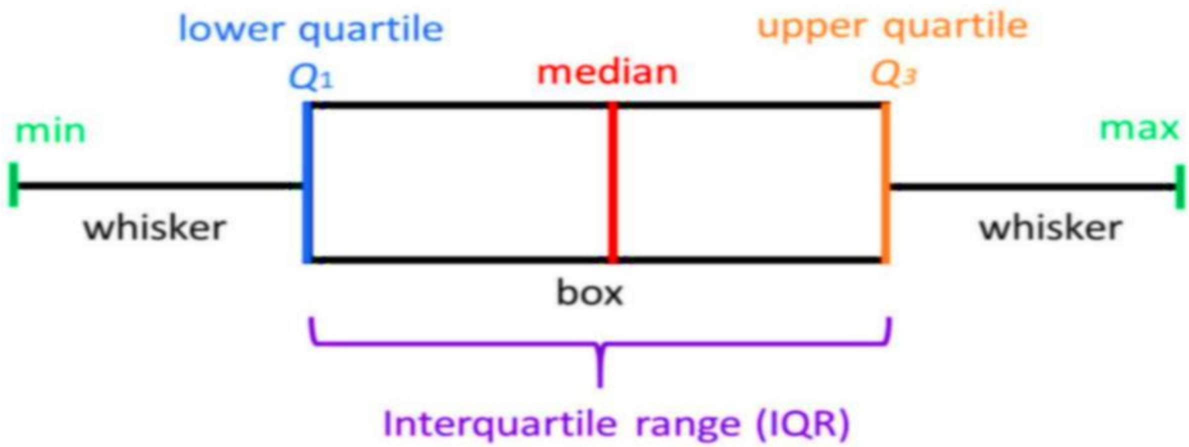
- Data is divided into equal intervals called **bins**.
- Each bar represents one interval.
- The **height of the bar** indicates the frequency (number of observations).
- There are **no gaps** between bars because the data is continuous.
- The width of each bar usually remains the same.
- The x-axis represents class intervals.
- The y-axis represents frequency.

Quartile Plot (Box Plot)

A **Box Plot** (also called a **Quartile Plot**) is a graph that summarizes a dataset using the **five-number summary**.

Five-Number Summary

- Minimum
- First Quartile (Q1)
- Median (Q2)
- Third Quartile (Q3)
- Maximum



Features

- The rectangular box represents the **Interquartile Range (IQR)**.
- A line inside the box represents the **median**.
- Whiskers extend from the box to the minimum and maximum values (excluding outliers).
- Outliers are displayed as separate points.

Important Terms

Minimum

- Smallest value in the dataset.

Maximum

- Largest value in the dataset.

Median (Q2)

- Middle value of the ordered dataset.
- Divides data into two equal halves.

First Quartile (Q1)

- 25% of data lies below Q1.
- 75% lies above Q1.

Third Quartile (Q3)

- 75% of data lies below Q3.
- 25% lies above Q3.

Interquartile Range (IQR)

- Formula:
- **IQR = Q3 − Q1**
- It measures the spread of the middle 50% of the data.

Whiskers

- Lines extending from the box.
- Connect the box to the smallest and largest values (excluding outliers).
- Show the spread of the data.

Outliers

- Outliers are unusually high or low values compared to the rest of the dataset.

Outliers

Outliers are unusually high or low values.

They are calculated using:

Lower Limit

$$Q1 - 1.5(IQR)$$

Upper Limit

$$Q3 + 1.5(IQR)$$

Using the boxplot, find the following: 65 70 75 80 85 90 95 100

Find:

a) Minimum value

b) Maximum value

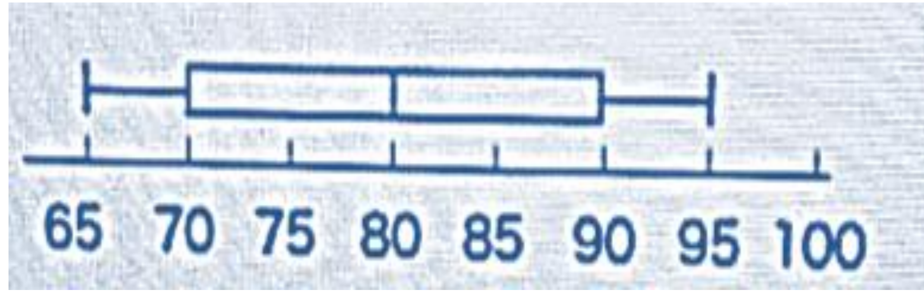
c) First quartile

d) Second quartile

e) Third quartile

f) Interquartile range

g) Median



Minimum (left whisker) = 65

First Quartile Q1 (left edge of box) = 70

Median / Second Quartile Q2 (line inside box) = 80

Third Quartile Q3 (right edge of box) = 90

Maximum (right whisker) = 95

Calculation of IQR

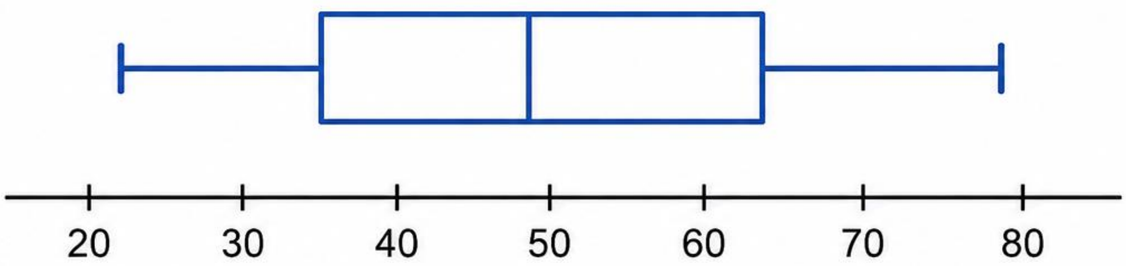
$$IQR = Q_3 - Q_1 = 90 - 70 = 20$$

TUTORIAL

Question 1:

Using the boxplot, find the following:

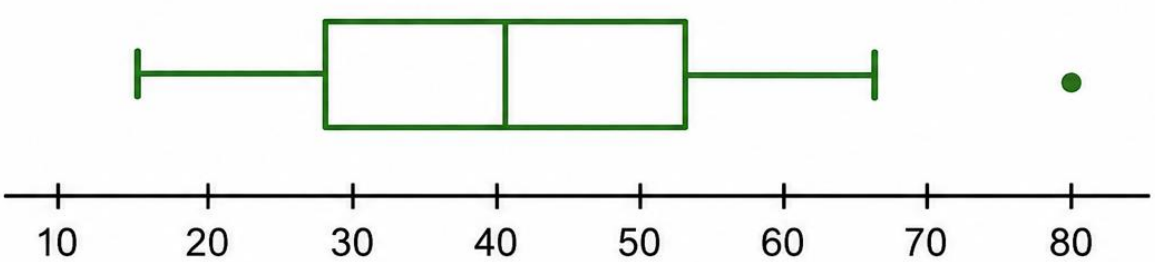
a) Minimum value
b) Maximum value
c) First quartile
d) Second quartile
e) Third quartile
f) Interquartile range
g) Median
h) Outlier



Question 2:

Using the boxplot, find the following:

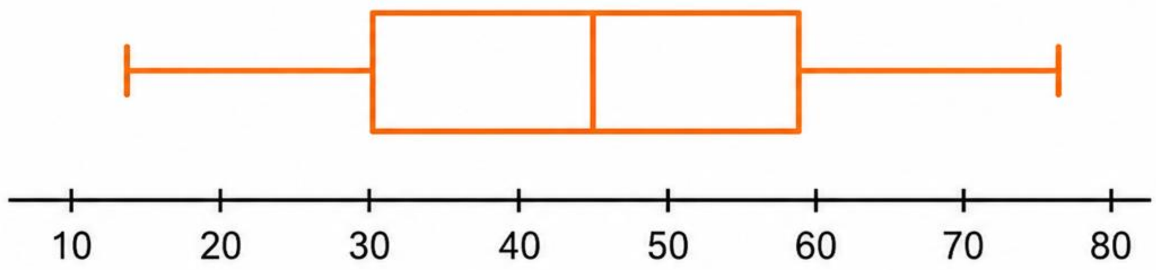
a) Minimum value
b) Maximum value
c) First quartile
d) Second quartile
e) Third quartile
f) Interquartile range
g) Median
h) Outlier



Question 3:

Using the boxplot, find the following:

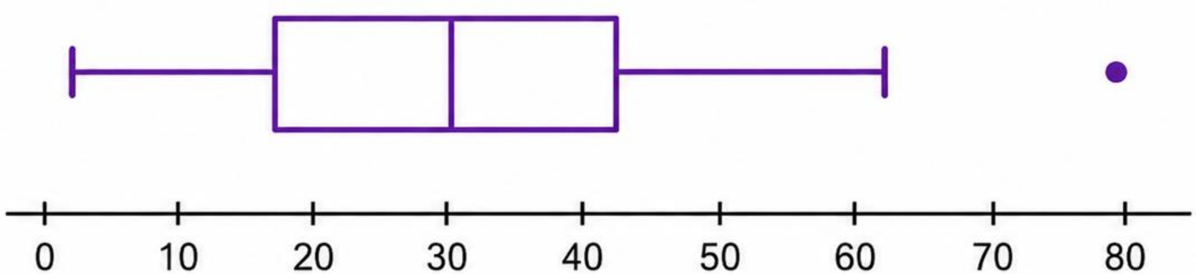
a) Minimum value
b) Maximum value
c) First quartile
d) Second quartile
e) Third quartile
f) Interquartile range
g) Median
h) Outlier



Question 4:

Using the boxplot, find the following:

a) Minimum value
b) Maximum value
c) First quartile
d) Second quartile
e) Third quartile
f) Interquartile range
g) Median
h) Outlier

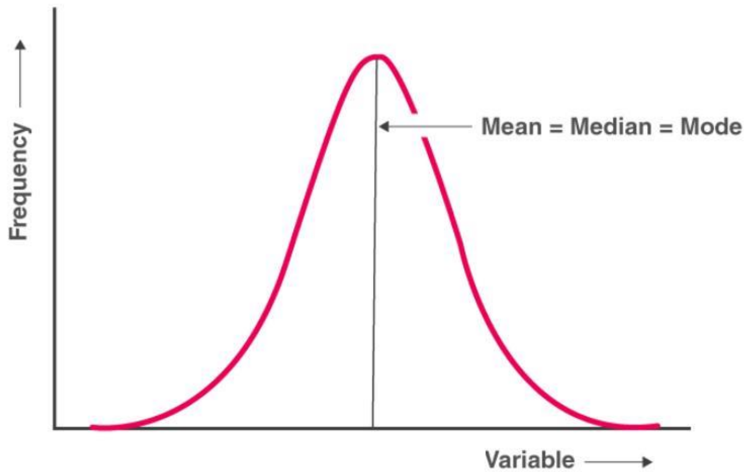


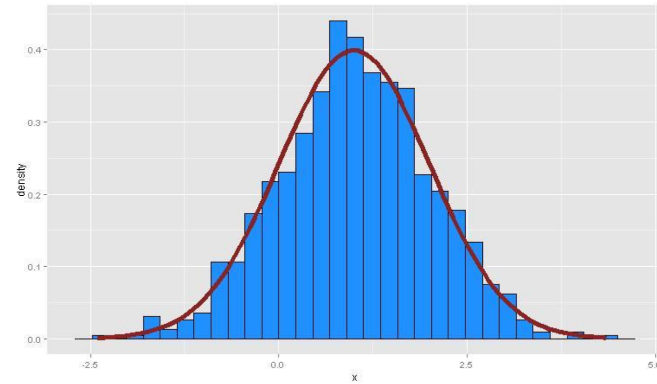
Distribution Chart

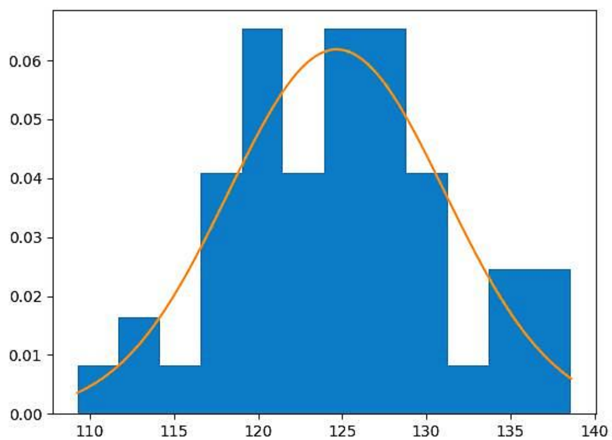
- A **Distribution Chart** shows how data values are distributed across different ranges.
- It helps understand the overall pattern of the dataset.

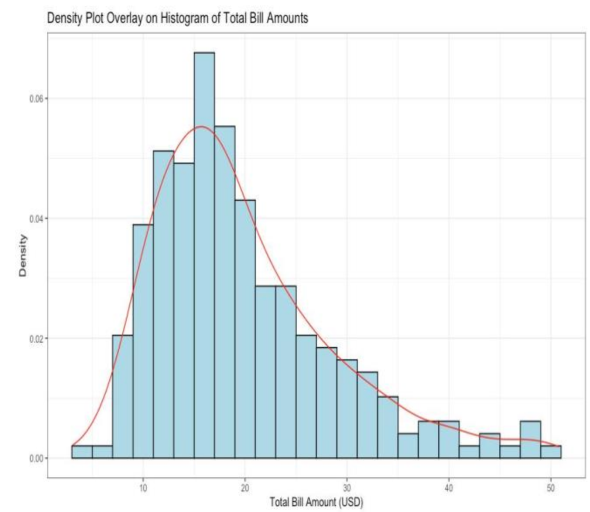
Features

- Displays data distribution.
- Shows spread of data.
- Helps identify skewness.


Frequency
Mean = Median = Mode
Variable


density
0.4
0.3
0.2
0.1
0.0
-2.5
0.0
2.5
5.0
x


0.06
0.05
0.04
0.03
0.02
0.01
0.00
110
115
120
125
130
135
140

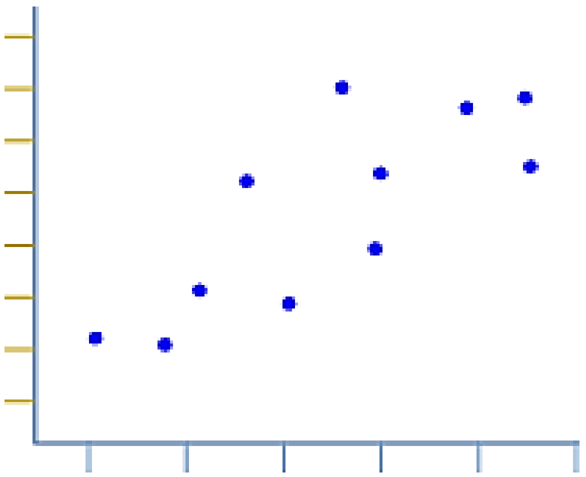

Density Plot Overlay on Histogram of Total Bill Amounts
Density
0.06
0.04
0.02
0.00
10
20
30
40
50
Total Bill Amount (USD)

Scatter Plot

- A **Scatter Plot** is a graph used to display the relationship between two numerical variables.
- Each observation is represented by one point.

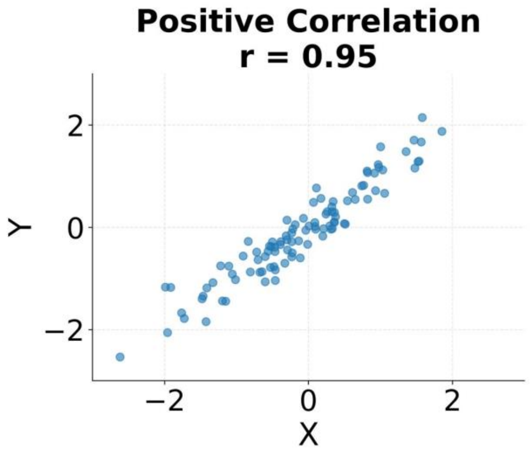
Features

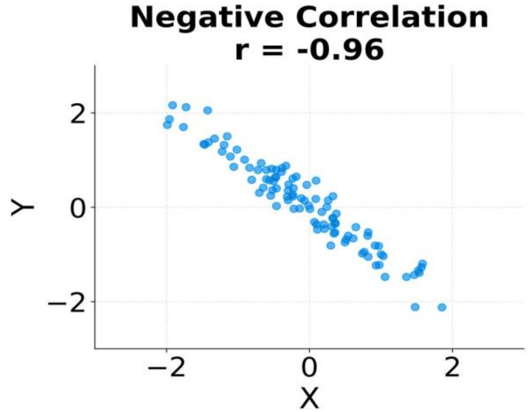
- Every point has x and y coordinates.
- Shows relationships between variables.
- Detects trends.
- Identifies outliers.
- Helps estimate correlation.

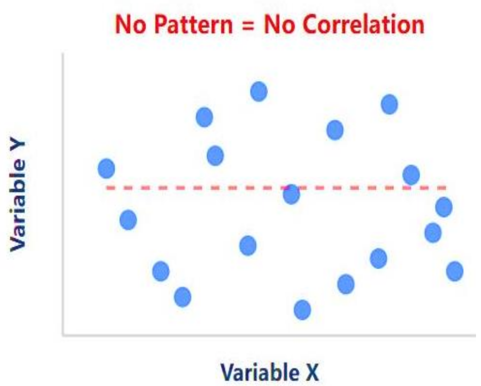


Types of Correlation

- **Positive Correlation**
  - Both variables increase together.
  - Example:
    - Study hours ↑
    - Marks ↑
- **Negative Correlation**
  - One variable increases while the other decreases.
  - Example:
    - Speed ↑
    - Travel time ↓
- **No Correlation**
  - No relationship exists.
  - Data points appear randomly scattered.

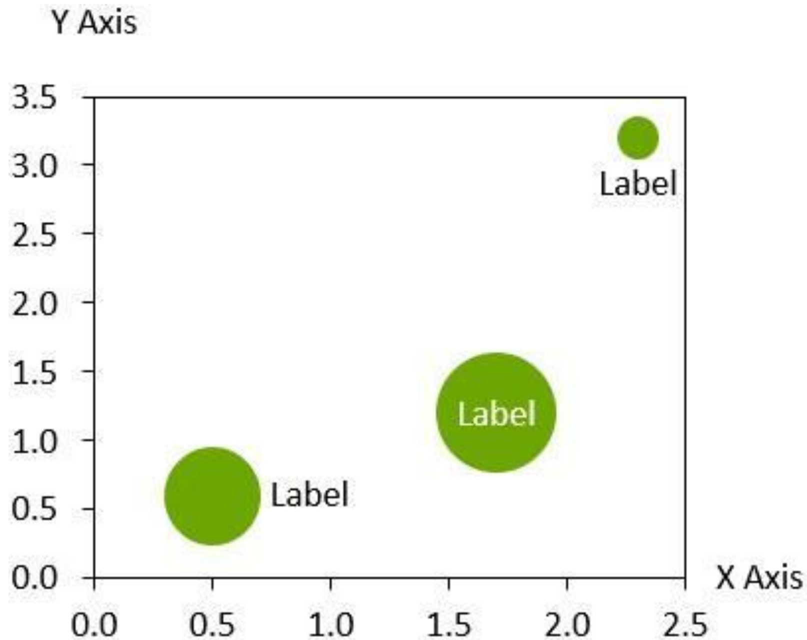






Bubble Chart

- A **Bubble Chart** is an extension of a scatter plot in which each data point is represented by a **bubble**.
- The position of the bubble is determined by the **X-axis** and **Y-axis**, while the **size of the bubble** represents a third numerical variable.



Advantages

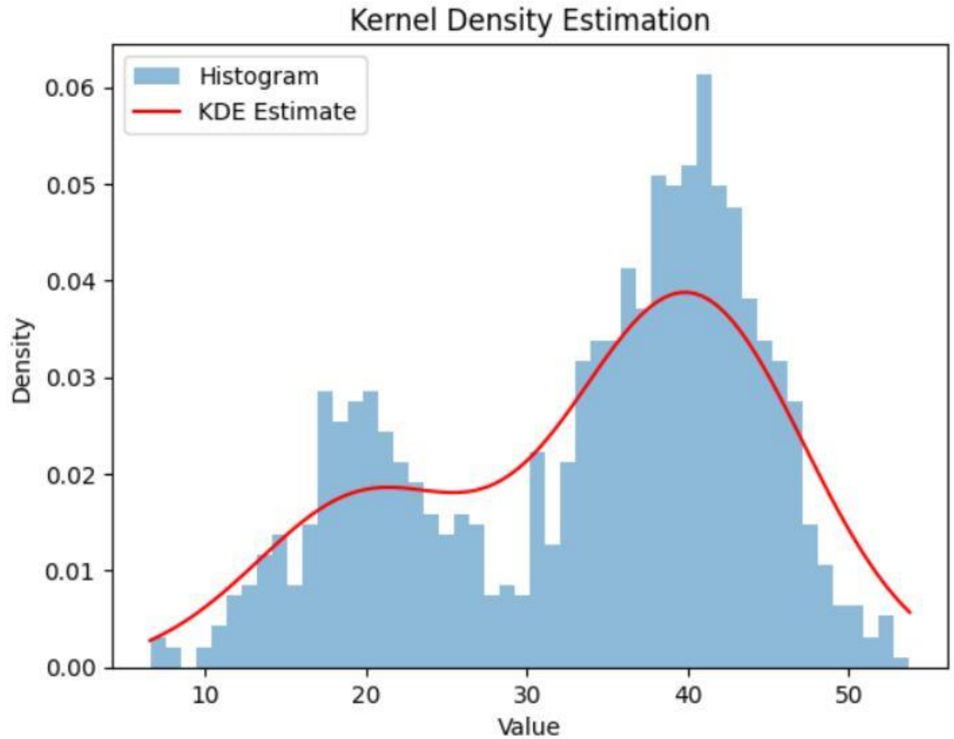
- Represents three variables in a single graph.
- Makes comparison between observations easier.
- Reveals clusters, trends, and outliers.
- Provides more information than a scatter plot.
- Useful for decision-making and business analysis.

Disadvantages

- Bubble overlap can reduce readability.
- Difficult to compare bubble areas accurately.
- Unsuitable for very large datasets with many observations.
- Small bubbles may become difficult to notice.

Density Chart

- A **Density Chart**, also known as a **Density Plot**, is a statistical graph that illustrates the distribution of continuous data using a smooth probability density curve.
- Unlike a histogram, which groups data into bars, a density plot estimates the **probability density function** using a smoothing technique called **Kernel Density Estimation (KDE)**.



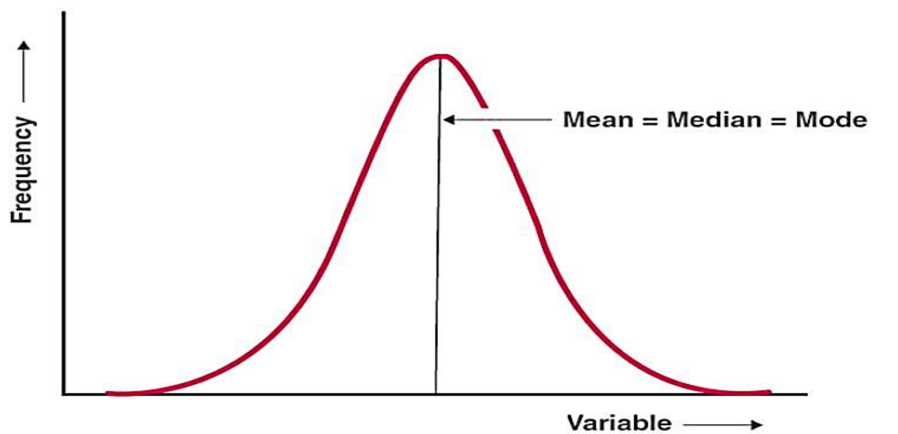
Types of Density Distributions

1. Normal Distribution (Symmetric Distribution)

A **Normal Distribution** is the most common type of probability distribution.

It has a **bell-shaped, symmetric curve**, where data values are evenly distributed around the center.

In a normal distribution, the left and right sides of the curve are mirror images of each other.



Characteristics

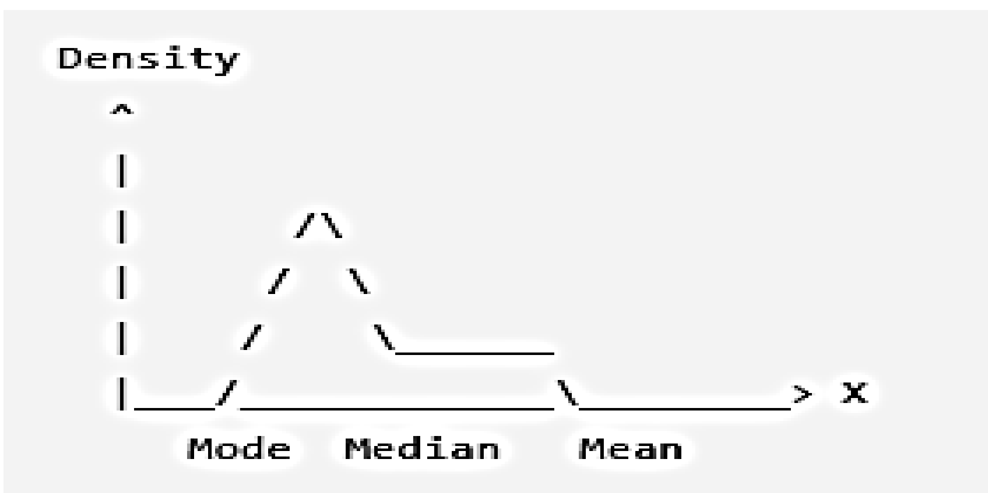
- Bell-shaped and symmetric.
- Mean = Median = Mode.
- Highest frequency occurs at the center.
- Frequencies decrease equally on both sides.

2. Right-Skewed Distribution (Positive Skew)

A **Right-Skewed Distribution** (Positive Skew) occurs when most observations are concentrated on the left side of the graph, while a few unusually large values create a **long tail toward the right**.

Characteristics

- Long tail toward higher values (right side).
- Most observations have relatively small values.

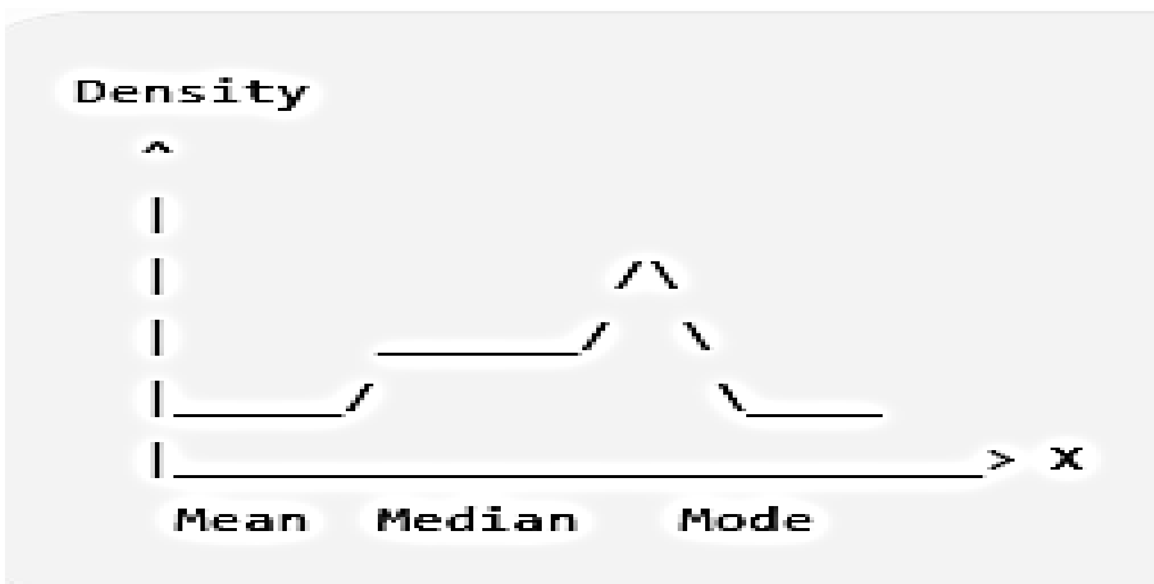


3. Left-Skewed Distribution (Negative Skew)

A **Left-Skewed Distribution** (Negative Skew) occurs when most observations are concentrated on the right side, while a few unusually small values create a **long tail toward the left**.

Characteristics

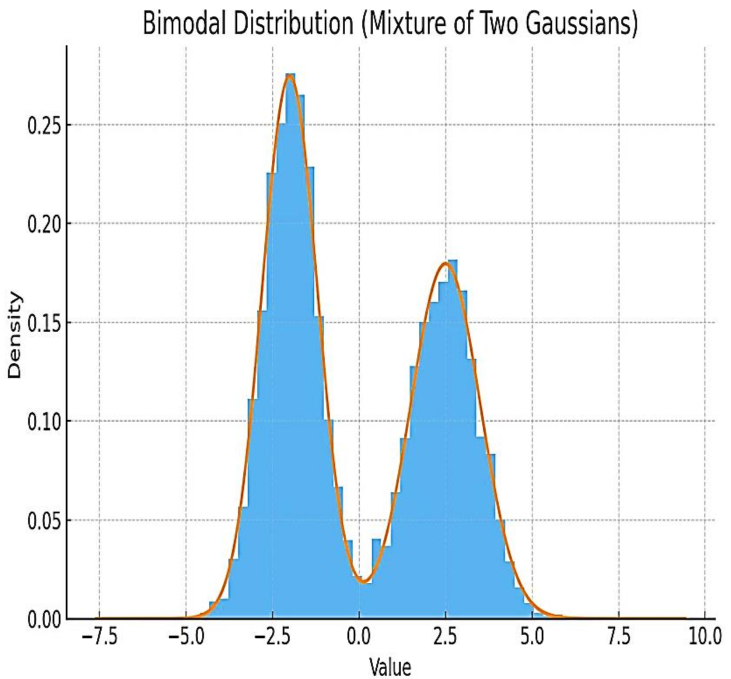
- Long tail toward lower values (left side).
- Most observations have relatively large values.



4. Bimodal Distribution

A **Bimodal Distribution** is a distribution that contains **two distinct peaks (modes)**.

These peaks indicate that the dataset is made up of **two different groups or populations**.

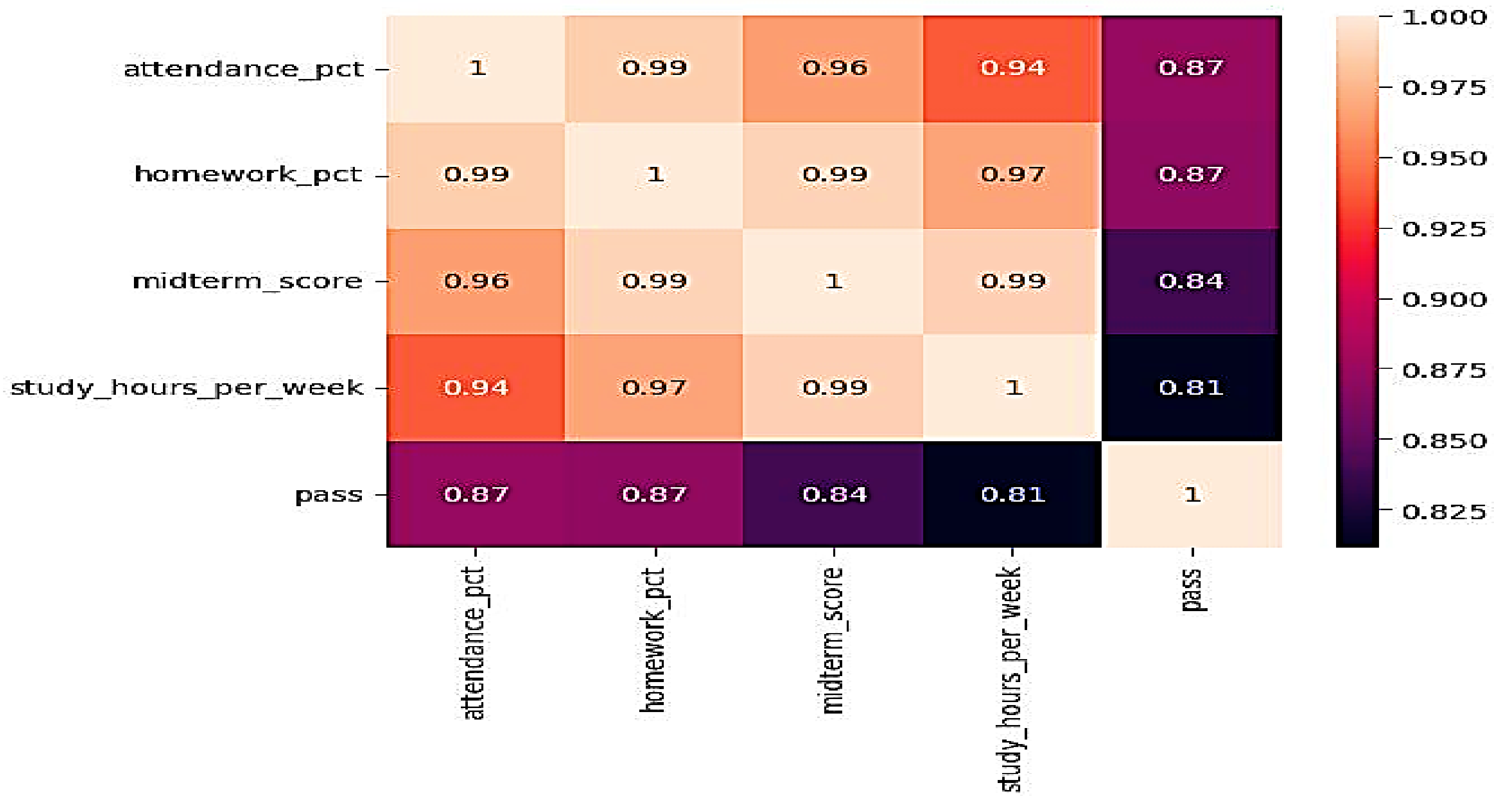


Characteristics

- Two prominent peaks.
- Indicates two modes.
- May represent two different populations.
- Often observed when combining datasets with different characteristics.

Heatmap

- A **Heatmap** is a graphical representation of numerical data in which individual values are represented by different colors.
- It enables users to identify patterns, trends, clusters, and relationships quickly without examining individual numbers.
- Heatmaps are especially useful in **machine learning**, **data mining**, and **statistical analysis** for visualizing correlation matrices and large datasets.



Components

- **Rows:** Variables or observations.
- **Columns:** Variables or categories.
- **Cells:** Numerical values.
- **Color Scale:** Represents the magnitude of values.
- **Color Legend:** Explains the meaning of colors.

Working Principle

- Each cell in the matrix is assigned a color based on its numerical value.
- ***Dark colors*** generally indicate **higher values.**
- ***Light colors*** represent **lower values.**
- Different color gradients help distinguish positive and negative values in correlation matrices.

Types of Heatmaps

Correlation Heatmap

- Shows relationships among variables.

Cluster Heatmap

- Groups similar observations and variables.

Geographic Heatmap

- Displays intensity based on geographical locations.

Matrix Heatmap

- Represents numerical values within a matrix.

Visualization	Number of Variables Displayed	How Variables Are Represented	Purpose
Histogram	1 variable	X-axis = Continuous variable, Y-axis = Frequency	Shows the frequency distribution of a continuous variable.
Box Plot (Quartile Plot)	1 variable (or multiple groups for comparison)	Displays minimum, Q1, median, Q3, maximum, and outliers	Summarizes the distribution, spread, and outliers of a variable.
Distribution Chart (Distribution Plot)	1 variable	X-axis = Variable values, Y-axis = Frequency or Probability	Shows how the values of a variable are distributed. It may appear as a histogram, density curve, or frequency distribution.
Scatter Plot	2 variables	X-axis = Variable 1, Y-axis = Variable 2	Shows the relationship or correlation between two numerical variables.
Bubble Chart	3 variables (4 if color is used)	X-axis = Variable 1, Y-axis = Variable 2, Bubble Size = Variable 3	Shows relationships among three variables simultaneously.
Density Chart	1 variable	X-axis = Continuous variable, Y-axis = Probability Density	Displays the probability distribution of a continuous variable using a smooth curve.
Heatmap	Multiple variables	Rows = Variables, Columns = Variables, Cell Color = Value	Shows relationships, correlations, or intensity values among multiple variables.


THE
END